

# Find Sum of Numbers in a String

## Java Code

```
class Solution {  
    public static int findSum(String s) {  
  
        char[] arr = s.toCharArray();  
  
        int sum = 0;  
        int num = 0;  
  
        for (int i = 0; i < arr.length; i++) {  
  
            if (Character.isDigit(arr[i])) {  
                num = num * 10 + (arr[i] - '0');  
            }  
            else {  
                sum = sum + num;  
                num = 0;  
            }  
        }  
  
        // Add the last number  
        sum = sum + num;  
  
        return sum;  
    }  
}
```

---

## Visualization

### Input

```
s = "abc12de3fg45"
```

Characters:

a b c 1 2 d e 3 f g 4 5  
↑ ↑ ↑ ↑ ↑

### Step 1 — 1

```
num = 0 * 10 + 1  
      = 1
```

```
sum = 0  
num = 1
```

### Step 2 — 2

```
num = 1 * 10 + 2  
      = 12
```

```
sum = 0  
num = 12
```

The multiplication by 10 joins digits:

```
1 → 1 × 10 + 2 → 12
```

### Step 3 — d

d is not a digit:

```
sum = sum + num  
      = 0 + 12  
      = 12
```

Then:

```
num = 0
```

Visualization:

```
"12" → sum  
    ↓  
    sum = 12  
  
num → 0
```

---

#### Step 4 — 3

```
num = 0 * 10 + 3  
    = 3
```

```
sum = 12  
num = 3
```

---

#### Step 5 — f

```
sum = 12 + 3  
    = 15  
  
num = 0
```

---

#### Step 6 — 4

```
num = 0 * 10 + 4  
    = 4
```

---

#### Step 7 — 5

```
num = 4 * 10 + 5  
    = 45
```

Current:

```
sum = 15  
num = 45
```

---

## Last Number

The loop ends after reading `45`.

There is no non-digit character after it to execute:

```
sum = sum + num;
```

So we add the last number manually:

```
sum = 15 + 45  
      = 60
```

## Final Output

```
60
```

---

## Complete Flow

```
"abc12de3fg45"
```

↓

```
abc → ignore
```

↓

```
1 → num = 1
```

```
2 → num = 12
```

↓

```
d → sum = 12, num = 0
```

↓

```
3 → num = 3
```

↓

```
f → sum = 15, num = 0
  ↓
4 → num = 4
5 → num = 45
  ↓
End → sum = 15 + 45
    ↓
    60
```

## Key Logic

```
Digit
  ↓
num = num * 10 + digit

Non-digit
  ↓
sum = sum + num
  ↓
num = 0
```

## Complexity

```
Time → O(n)
Space → O(n)
```

`O(n)` space includes the `char[]` created by `toCharArray()`.